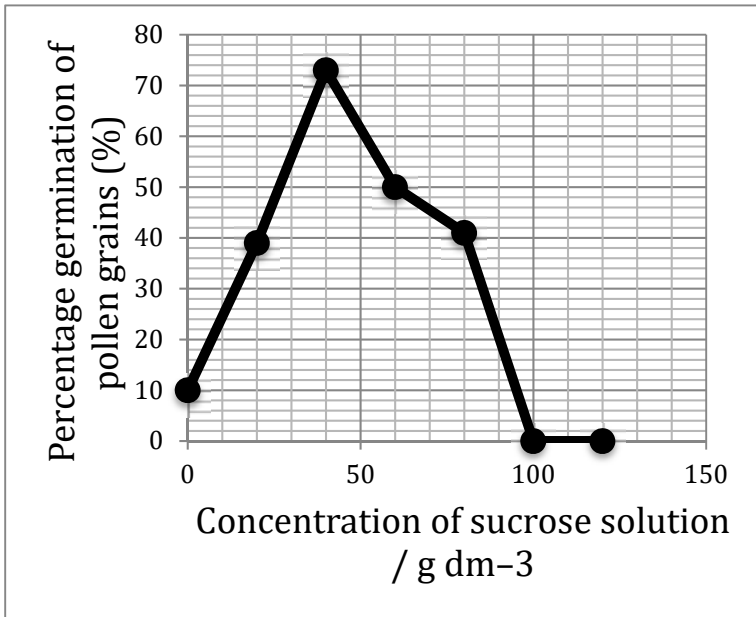


Question number	Answer	Additional guidance	Mark
1(a)(i)	concentration of sucrose (solution) (1)	Accept sucrose concentration Do not accept 'sucrose' without concentration	(1)

Question number	Answer	Additional guidance	Mark
1(a)(ii)	temperature / pH (1)	Accept light intensity	(1)

Question number	Answer	Additional guidance	Mark
1(a)(iii)	An explanation that includes the following points: - an effect of the factor on the germination (1) - because enzymes are involved in (pollen germination/metabolism) (1) - description of {metabolism/enzymes} being affected by the factor (1)	Accept growth Do not accept reference to seeds	(3)

Question number	Answer	Additional guidance	Mark																
1(b)(i)	A graph showing the following features: • A axes the right way round (x = sucrose conc., y = percentage germination) and linear scale (1) • L axes correctly labelled with units (x - sucrose concentration, g dm⁻³, y - percentage germination of pollen grains (1) • P correct plotting of all values (1) • S line joining points accurately ruled (1)	 <table><caption>Data points from the graph</caption><thead><tr><th>Concentration of sucrose solution / g dm⁻³</th><th>Percentage germination of pollen grains (%)</th></tr></thead><tbody><tr><td>0</td><td>10</td></tr><tr><td>20</td><td>40</td></tr><tr><td>40</td><td>75</td></tr><tr><td>60</td><td>50</td></tr><tr><td>80</td><td>40</td></tr><tr><td>100</td><td>0</td></tr><tr><td>120</td><td>0</td></tr></tbody></table>	Concentration of sucrose solution / g dm ⁻³	Percentage germination of pollen grains (%)	0	10	20	40	40	75	60	50	80	40	100	0	120	0	(4)
Concentration of sucrose solution / g dm ⁻³	Percentage germination of pollen grains (%)																		
0	10																		
20	40																		
40	75																		
60	50																		
80	40																		
100	0																		
120	0																		

Question number	Answer	Additional guidance	Mark
1(b)(ii)	An explanation that includes the following points: - repeat investigation using more sucrose concentrations (1) - between 20 and 60 g dm⁻³ (1)		(2)

Question number	Answer	Additional guidance	Mark
1(c)(i)	A calculation showing the following steps: - selection of correct values from the table (1) - calculation of percentage (1)	166 – 78 / 88 Example of calculation: $(88 \div 78) \times 100 = 112.82 \%$ Accept 113/112.8/112.821 Correct answer with no working gains both marks	(2)

Question number	Answer	Additional guidance	Mark
1(c)(ii)	An answer that includes five of the following points: • use a range of at least 5 concentrations of (sucrose) solutions (1) • detail of dilution method using the sucrose solution (of 500 g dm^{-3}) (1) • at least one control variable named (1) • use 100 ppm boric acid solution (1) • use a microscope and graticule to measure pollen tube length (1) • stated times (for measurements of pollen tube) (1)	Accept stock and distilled water as one of the concentrations.	(5)